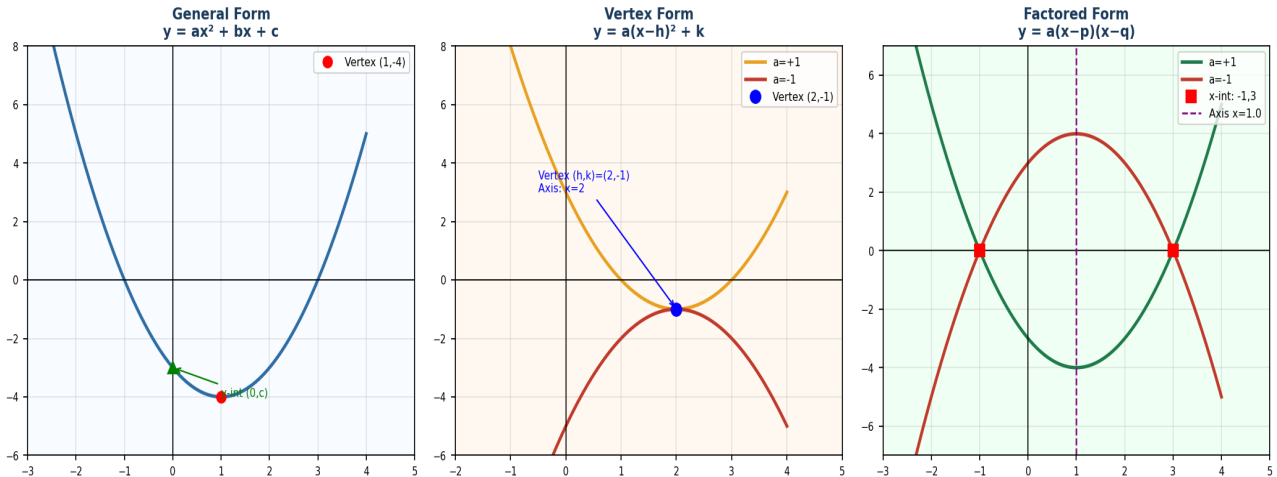


Parabolas & Non-Linear Curves

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1. The Three Forms of a Parabola



Form	Equation	What you can read off directly
General	$y = ax^2 + bx + c$	y-intercept = c; use formulas for everything else
Vertex (Turning Point)	$y = a(x - h)^2 + k$	Vertex (h, k); axis of symmetry $x = h$; direction of opening
Factored (x-intercept)	$y = a(x - p)(x - q)$	x-intercepts at $x = p$ and $x = q$; axis $x = (p+q)/2$

2. General Form $y = ax^2 + bx + c$

Key Features

- **y-intercept:** set $x = 0 \rightarrow y = c \rightarrow$ point (0, c)
- **x-intercepts (roots):** set $y = 0$, solve $ax^2 + bx + c = 0$
- **Axis of symmetry:** $x = -b / (2a)$
- **Vertex (turning point):** $x = -b/(2a)$, then substitute back to find y
- **Direction:** $a > 0$ opens up (min); $a < 0$ opens down (max)

Discriminant $\Delta = b^2 - 4ac$

$\Delta > 0 \rightarrow$ two distinct real roots (crosses x-axis twice)

$\Delta = 0 \rightarrow$ one real root (touches x-axis — vertex on axis)

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Used to find x-intercepts from general form.

Completing the Square

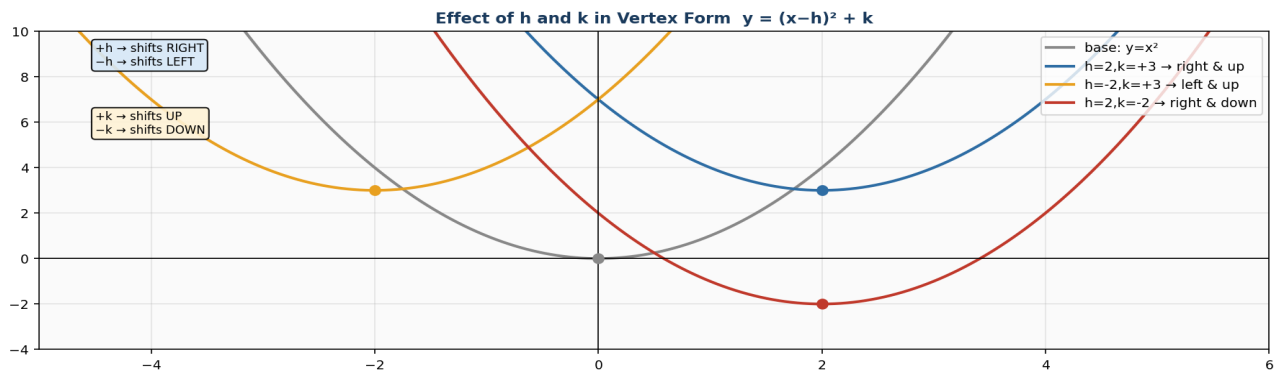
Convert general $\rightarrow$ vertex form:

1. $y = a(x^2 + (b/a)x) + c$
2. $y = a(x + b/(2a))^2 - b^2/(4a) + c$
3. Vertex: $h = -b/(2a)$, $k = c - b^2/(4a)$

$\Delta < 0 \rightarrow$ no real roots (parabola doesn't cross x-axis)

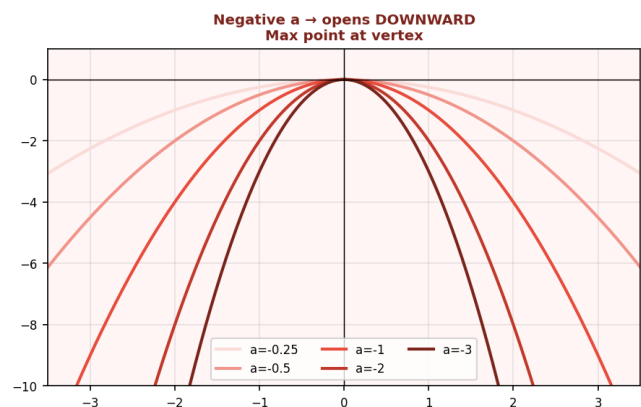
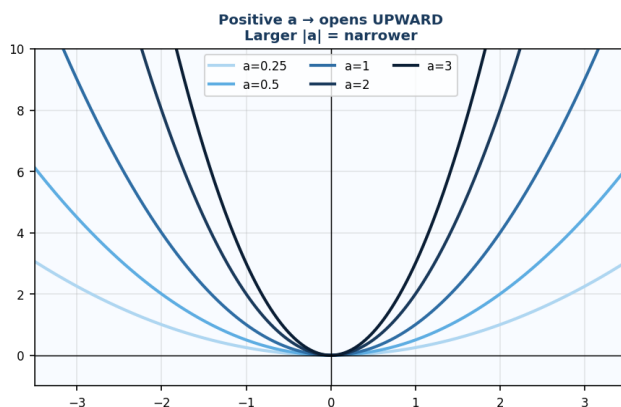
Converting $\rightarrow$ Factored Form

3. Vertex Form $y = a(x - h)^2 + k$



Feature	Rule	Example $y = 2(x-3)^2 - 5$
Vertex (turning point)	(h, k)	$(3, -5)$
Axis of symmetry	$x = h$	$x = 3$
y-intercept	Set $x = 0 \rightarrow y = ah^2 + k$	$y = 2(9) - 5 = 13 \rightarrow (0, 13)$
x-intercepts	Set $y=0 \rightarrow (x-h)^2 = -k/a \rightarrow x = h \pm \sqrt{-k/a}$	$x = 3 \pm \sqrt{(5/2)} \approx 1.42$ or 4.58
Min/Max value	$a>0$: min = k at $x=h$; $a<0$: max = k at $x=h$	$a=2>0 \rightarrow$ minimum value = -5
Domain	All real x (always)	$x \in \mathbb{R}$
Range	$a>0$: $y \geq k$; $a<0$: $y \leq k$	$y \geq -5$

4. Effect of Dilation Factor a



Value of a	Direction	Shape	Vertex is a...
$a > 1$	Opens UPWARD $\uparrow$	Narrower than $y=x^2$	MINIMUM point
$0 < a < 1$	Opens UPWARD $\uparrow$	Wider than $y=x^2$	MINIMUM point
$a = 1$	Opens UPWARD $\uparrow$	Standard width	MINIMUM point

$-1 < a < 0$	Opens DOWNWARD ↓	Wider than $y = -x^2$	MAXIMUM point
$a < -1$	Opens DOWNWARD ↓	Narrower than $y = -x^2$	MAXIMUM point

5. Factored Form $y = a(x - p)(x - q)$

Key Features

- **x-intercepts (roots):** $x = p$ and $x = q$
- **Axis of symmetry:** $x = (p + q) / 2$
- **Vertex x-coordinate:** same as axis $= (p+q)/2$
- **Vertex y-coordinate:** substitute axis x into equation
- **y-intercept:** set $x = 0 \rightarrow y = a(-p)(-q) = apq$

Note on sign convention

$y = a(x - p)(x - q)$ has roots at $+p$ and $+q$

$y = (x + 3)(x - 2) \rightarrow$ roots at $x = -3$ and $x = +2$

(because $x+3=0$ gives $x=-3$)

Worked Example

$$y = -2(x + 1)(x - 3)$$

$a = -2 \rightarrow$ opens downward, max vertex

x-intercepts: $x = -1$ and $x = 3$

Axis: $x = (-1+3)/2 = 1$

Vertex $x=1$: $y = -2(2)(-2) = 8 \rightarrow$ vertex $(1, 8)$

y-int: $x=0 \rightarrow y = -2(1)(-3) = 6 \rightarrow (0, 6)$

Expanding to General Form

Multiply out: $a(x-p)(x-q) = ax^2 - a(p+q)x + apq$

So: $b = -a(p+q)$ and $c = apq$

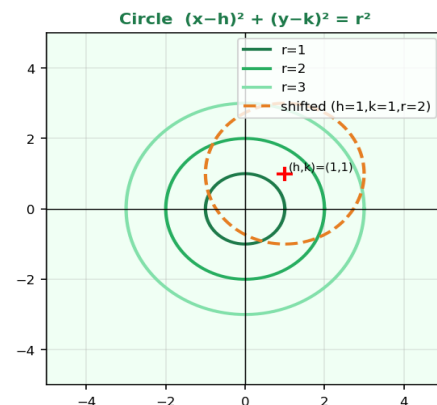
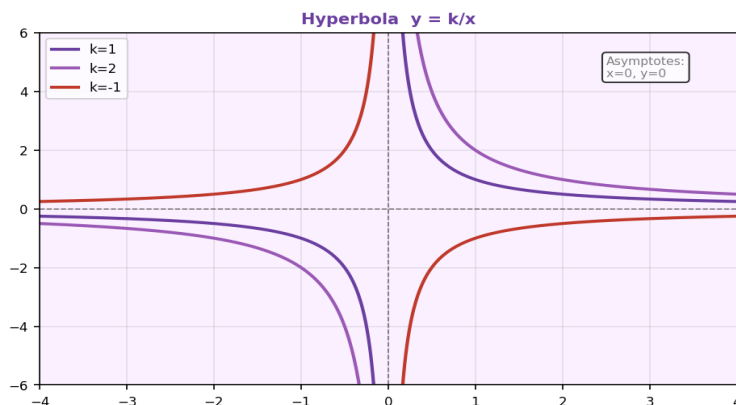
6. Converting Between Forms — Quick Reference

FROM	TO	Method
General	Vertex	Complete the square, OR use $h = -b/(2a)$ then $k = f(h)$
General	Factored	Factor the quadratic, or use quadratic formula to find roots p, q
Vertex	General	Expand: $a(x-h)^2 + k = ax^2 - 2ahx + ah^2 + k$
Vertex	Factored	Find roots from vertex form (set $y=0$), then write factored form
Factored	General	Expand: $a(x-p)(x-q) = ax^2 - a(p+q)x + apq$
Factored	Vertex	Use axis $x=(p+q)/2$ to find vertex, write as $a(x-h)^2+k$

7. Step-by-Step: Sketching Any Parabola

Step	Task	How
Step 1	Identify a	Determine sign (up/down) and size (narrow/wide) of parabola.
Step 2	Find y-intercept	Set $x = 0$. In general form: $y = c$. In factored: $y = a(-p)(-q)$.
Step 3	Find x-intercepts	Set $y = 0$. Factorise, quadratic formula, or read from factored form.
Step 4	Find axis of symmetry	$x = -b/(2a)$ OR $x = h$ OR $x = (p+q)/2$
Step 5	Find vertex	Substitute axis x into equation to get y . Vertex = (axis x , y).
Step 6	Determine range	$a > 0$: $y \geq k$ (min). $a < 0$: $y \leq k$ (max).
Step 7	Sketch	Plot y-int, x-ints, vertex. Draw smooth U (or $\cap$) through them.

8. Other Non-Linear Curves: Hyperbola & Circle



Hyperbola $y = k/x$	Feature
Equation	$y = k/x$ (also written $xy = k$)
Asymptotes	$x = 0$ (y-axis) and $y = 0$ (x-axis) — curve never crosses them
Quadrants	$k > 0$: Q1 & Q3 (both branches positive); $k < 0$: Q2 & Q4
Domain	$x \in \blacksquare, x \neq 0$
Range	$y \in \blacksquare, y \neq 0$
Symmetry	Symmetric about $y = x$ and $y = -x$
Shifted form	$y = k/(x-h) + c \rightarrow$ asymptotes at $x=h, y=c$

Circle $(x-h)^2 + (y-k)^2 = r^2$	Feature
Equation	$(x-h)^2 + (y-k)^2 = r^2$
Centre	(h, k)
Radius	$r = \sqrt{r^2}$ — always positive
x-intercepts	Set $y=0$: $(x-h)^2 = r^2 - k^2 \rightarrow x = h \pm \sqrt{r^2 - k^2}$ (if $r^2 \geq k^2$)
y-intercepts	Set $x=0$: $(y-k)^2 = r^2 - h^2 \rightarrow y = k \pm \sqrt{r^2 - h^2}$ (if $r^2 \geq h^2$)
Domain	$h - r \leq x \leq h + r$
Range	$k - r \leq y \leq k + r$
Special case	Centre at origin: $x^2 + y^2 = r^2$
Tip: General form	$x^2 + y^2 + Dx + Ey + F = 0 \rightarrow$ complete the square on x and y to get centre-radius form

9. Quick Tips & Memory Aids

Icon	Topic	Memory Aid
■	Axis of symmetry	Always halfway between the two x-intercepts. Axis $x = (p+q)/2$.

■	Quick y-intercept	ALWAYS just set $x = 0$. In general form, $y = c$ instantly.
■	Discriminant check	Before solving, check $\Delta = b^2 - 4ac$. Saves time — if $\Delta < 0$, no real roots!
$\leftrightarrow$	h shifts horizontally	In $y = a(x-h)^2 + k$, the sign of h is OPPOSITE to the shift direction. $x-3 \rightarrow$ right 3.
$\updownarrow$	k shifts vertically	k is straightforward: $+k$ up, $-k$ down.
■	Finding a from graph	Given vertex (h,k) and one other point (x,y) : $a = (y-k)/(x-h)^2$
==■	Two roots $\rightarrow$ factored	If you know x-intercepts p and q , write $y = a(x-p)(x-q)$ immediately.
■	Circle check	Make sure coefficients of x^2 and y^2 are both 1 and equal before using circle formula.